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BCSE415L

Human Computer Interaction

DIGITAL ASSIGNMENT-1

**Personalized Gesture Vocabulary Builder: A Few-Shot
Learning System for Augmentative and Alternative
Communication**

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1. Topic Finalization

1.1 Background

Our original plan was to create a webcam-based hand gesture control system. We wanted to use a few-shot learning approach to let users teach the system a new gesture with just a few examples. We relied on MediaPipe for hand landmark tracking and a lightweight neural network for recognition. Although this approach was technically sound, the way we planned to apply it—using gestures to control media playback, slides, or other general commands—turned out to be a popular choice for HCI course projects this term. Several other groups came up with similar ideas, and our topic was marked as too similar to existing submissions.

Instead of abandoning our technical core, we decided to keep it and change its application. The standout feature of our original system was the ability to enroll a new, personal gesture using just a few examples. However, we initially aimed to use this ability for a generic purpose, like skipping a slide or muting a call. We realized that this capability would be more valuable if focused on a less crowded and more meaningful issue: helping a nonverbal or minimally verbal person develop their own small vocabulary of hand signs. This includes gestures that are unique to them and not part of any standard sign language or symbol set.

1.2 Reframing the Problem

A fixed, predetermined vocabulary is typically the foundation of standard AAC (Augmentative and Alternative Communication) technologies like picture boards, symbol grids, or speech-generating systems. This is effective for common demands, but it does not support individual gestures, such as a child's sign for a sibling's name, a particular method of requesting water, or a private signal for discomfort or agony that a speech therapist has never recorded. An adult must typically manually add an image and a label in order to teach a new symbol to a fixed AAC system. A new gesture must be demonstrated five to ten times in order for our system to learn it.

Rebuilding the pipeline is not necessary for this reframing. The few-shot enrollment stage, the embedding network, and MediaPipe hand tracking remain unaltered. The evaluation criteria (how consistently it avoids false triggers and how usable the enrollment flow is for a caregiver, rather than raw accuracy on a benchmark), the output (a spoken or displayed message instead of a media control command), and the framing of the contribution (a personal AAC vocabulary builder, not another gesture-controlled media player) all change.

1.3 Finalized Topic Statement

This project creates a Personalized Gesture Vocabulary Builder, an AAC tool that enables a nonverbal or minimally verbal user, or a caregiver acting on their behalf, to teach a webcam-based system a personal hand gesture from a few demonstrations. The system will then be able to reliably recognize the gesture in everyday situations and respond with either an on-screen symbol or a spoken message.

1.4 Objectives

1. Create a vision-based pipeline that uses a regular laptop webcam to extract hand landmarks in real time without the need for wearables or specialized sensors.
2. To add a new personal gesture from about five to ten instances without retraining the entire model, implement a few-shot enrollment procedure.
3. Instead of always imposing a best guess, reject inputs that do not clearly match any enrolled gesture to reduce false triggers.
4. In order to prevent learning gesture number twelve from subtly disrupting gesture number three, keep the detection of previously enrolled gestures consistent over time as new gestures are added.
5. To make the product a real communication tool rather than merely a classifier demo, pair each gesture with a spoken output and a straightforward on-screen display.
6. Evaluate the system for real-time responsiveness and for ease of use during the enrollment process, since a caregiver or the user themselves will be the one teaching the system.

1.5 Scope and Limitations

This is a course-level prototype, not a certified assistive device. It is designed for a single user session at a time on a laptop with a standard front-facing camera. It targets static and simple hand poses in this phase, with dynamic (motion-based) gestures identified as a natural extension. Message output is text and synthesized speech in English for this phase. The system is not intended to replace clinical AAC assessment or professional speech and language therapy, and any real deployment with an actual user would need to involve a speech-language pathologist.

2. Review of Related Literature

The literature below is organized into six groups: vision-based hand tracking, few-shot and prototype-based learning, few-shot sign language recognition, AAC foundations, gesture-based assistive interaction (the work closest to this project), and real-time stability and evaluation methodology.

2.1 Vision-Based Hand Tracking as the Input Layer

Zhang et al. (2020) describe MediaPipe Hands, a two-stage pipeline consisting of a palm detector and a hand landmark model that predicts twenty-one keypoints per hand from a single RGB camera in real time on mobile-class hardware. This is the foundation of our input layer: it lets us skip building a hand detector from scratch and instead work directly with a compact, well-tested landmark representation.

Because hand gesture recognition can be built on many different kinds of hardware, from depth cameras to wearable gloves, it was important to confirm that a plain webcam is a reasonable choice for an AAC tool that should be cheap and easy to set up. Oudah, Al-Naji, and Chahl (2020) review the wider landscape of gesture recognition techniques, comparing sensor-based and vision-based approaches, and note that vision-based methods avoid the burden of wearing extra hardware, which matters for a population that may already be managing other assistive devices.

Cui, Sunar, and Su (2025) provide a recent review of real-time, vision-based hand gesture recognition, summarizing how convolutional networks, recurrent networks, and attention-based models are used at each stage of the recognition pipeline (detection, tracking, feature extraction, classification). Their comparison across seven years of work helped confirm that a light embedding network on top of landmark coordinates, rather than a heavy end-to-end image classifier, is a reasonable and current design choice for a system meant to run on ordinary laptop hardware. Tripathi and Verma's survey on vision-based dynamic hand gesture recognition covers the broader family of temporal approaches (leap motion, recurrent networks, skeleton-based methods), which we draw on when discussing the dynamic gesture extension in our future work.

For the visual embedding branch of our architecture, we use MobileNetV2 as described by Sandler, Howard, Zhu, Zhmoginov, and Chen (2018). Its inverted residual blocks with linear bottlenecks were designed specifically to give strong accuracy at a low computational cost, which matters here because our system needs to run inference every frame on a standard laptop CPU without a dedicated GPU.

2.2 Few-Shot and Prototype-Based Recognition

The central mechanism of our redesigned system comes from Snell, Swersky, and Zemel (2017), who introduce Prototypical Networks. Rather than fine-tuning a classifier's weights every time a new class is added, a prototypical network learns an embedding space once, and then represents each class simply as the average (the "prototype") of its examples in that space. A new input is classified by finding its nearest prototype. This is exactly the mechanism we now use for gesture enrollment: showing five to ten examples of a new personal gesture produces one new prototype vector, added to a lookup table, with no gradient updates to the underlying network at all.

Wang, Yao, Kwok, and Ni (2020) provide a broad survey of few-shot learning approaches, covering metric-based methods like prototypical networks alongside optimization-based and data-augmentation-based alternatives. Their taxonomy helped us justify choosing a metric-based, prototype-averaging approach over a fine-tuning approach for this specific setting, since our users add gestures one at a time in a live session, and a training loop that has to run every time is a poor fit for that.

The reason this choice matters is directly tied to a known failure mode of fine-tuning based approaches. Kirkpatrick et al. (2017) describe catastrophic forgetting, the tendency of a neural network to lose previously learned information when it is updated on new data. Parisi, Kemker, Part, Kanan, and Wermter (2019) review this problem at length in the context of continual and lifelong learning systems. For our project, this is the direct justification for freezing the embedding backbone after initial training and only ever updating the prototype table: if we fine-tuned the network every time a new personal gesture was enrolled, there would be a real risk that teaching the system gesture number fifteen would degrade its recognition of gesture number two, which is unacceptable in a communication tool where every gesture may carry real meaning to the user.

Finally, Scheirer, Rocha, Sapkota, and Boulton (2013) formalize open set recognition, the idea that a classifier should be able to say "none of the above" rather than always forcing an input into one of its known categories. We use a similarity threshold on the distance to the nearest prototype so the system can reject an ambiguous or accidental hand position instead of guessing, which is critical for an AAC tool since a false positive means the system says something the user did not mean to say.

2.3 Few-Shot Sign Language Recognition

The closest technical relatives to our project are found in few-shot sign language recognition, since this field also deals with recognizing hand-based communication from very little labeled data per class.

Bohacek and Hruz (2023) demonstrate few-shot sign language recognition using existing online dictionary videos, showing that a handful of examples per sign class is workable even for large real-world sign vocabularies. Al Abdullah, Amoudi, and Alghamdi (2024) provide a comprehensive review of the sign language recognition field and repeatedly identify data scarcity as the central obstacle facing the field, which supports our decision to build around a few-shot design from the start rather than assuming a large labeled dataset will exist for every user's personal gestures. Bilge, Cinbis, and Ikizler-Cinbis (2023) push this even further with zero-shot sign language recognition, inferring unseen signs from learned attributes; we treat this as a harder, more research-oriented alternative to the simpler enrollment-based approach we have chosen, since our priority is something a caregiver can operate without machine learning expertise.

Wang and colleagues describe a "cornerstone network," a metric-based few-shot model for Chinese natural sign language that pairs a feature extractor with prototype-style matching, a design very close to the landmark stream in our own architecture. Kasapbaşı, Elbushra, Al-Hardanee, and Yilmaz (2022) present DeepASLR, a convolutional network based interface for American Sign Language recognition aimed specifically at hearing-impaired users, a concrete example of a full assistive pipeline built end to end. Hofman-Zweig and Ben-Assuli (2026), published in the journal *Technology and Disability*, present a contrastive-learning based sign recognition method that explicitly "requires no retraining for new signs," which is the closest published precedent we found for the enroll-once, never-retrain behavior we are designing toward.

2.4 Augmentative and Alternative Communication Foundations

To make sure the system is grounded in actual AAC practice and not just a machine learning exercise wearing an AAC label, we reviewed foundational work by Janice Light and David McNaughton. Light and McNaughton (2013) argue that AAC technology design should start from the person's communicative needs rather than from what the technology happens to do well, a principle that directly shaped our decision to let the system learn the user's gesture rather than requiring the user to learn a fixed gesture set. Light and McNaughton (2014) further define communicative competence for AAC users across linguistic, operational, social, and strategic

dimensions, which informed how we think about evaluating our tool beyond raw recognition accuracy.

Light, McNaughton, and Caron (2019) review emerging AAC technologies and the research directions the field considers important, and Light, Wilkinson, Thiessen, Beukelman, and Koch Fager (2019) provide design guidance specifically for AAC displays, both of which shaped our thinking on the output side of the system (the message board and spoken output), not just the gesture recognition side. Koch Fager, Fried-Oken, Jakobs, and Beukelman (2019) review access technologies for adults with complex communication needs and severe motor impairments, which broadened our sense of who this tool could serve beyond nonverbal children, including adults with conditions such as ALS or cerebral palsy who may retain only a small, reliable set of movements.

Mavritsakis (2024) documents a case study of a nonverbal teenager with autism spectrum disorder transitioning from a letter board to an iPad-based AAC system, illustrating in concrete terms how a personalized communication tool can open up a person's ability to express complex ideas over time. This case grounded our motivation in a real, human outcome rather than an abstract technical goal.

2.5 Gesture-Based Assistive Interaction (Closest Prior Systems)

The work that overlaps most directly with our project comes from Ascari, Pereira, and Silva. Their 2019 paper introduces PGCA, a computer vision system that lets people with motor and speech impairments create a personalized gestural interaction for AAC purposes, using motion representations such as Motion History Images together with SVM and CNN-based classifiers with transfer learning. Their 2020 follow-up, published in ACM Transactions on Accessible Computing, expands this methodology and evaluates it with both volunteers without disabilities and volunteers with motor and speech impairments. This is the closest existing system to what we are building, and it is worth being explicit about how our approach differs: their system generally retrains or fine-tunes a classifier per gesture set, while ours uses frozen embeddings and prototype averaging so that enrollment requires no training step at all, only an averaging operation, which should make enrollment faster and safer against forgetting previously taught gestures.

Colli Alfaro and Trejos (2022) study user-independent hand gesture classification using sensor fusion, which is a useful point of contrast: their goal is a model that works well across different people without personalization, while our system is deliberately person-dependent by design, since the whole point is to let one specific user's own idiosyncratic gesture be recognized reliably for them specifically.

Rafiq, Shi, and Albert (2024) present a wearable sensor-based few-shot continual learning framework for hand gestures aimed at motor-impaired individuals, explicitly motivated by the fact that "motor-impaired individuals require new gestures tailored to each individual's gesture motion and style." Although their signal comes from wearable accelerometers rather than a camera, their core design pattern, a replay-based few-shot continual learning framework built around a preserved latent embedding, cross-validates the general approach we are taking in a different sensing modality and reinforces that personalization, not generic recognition accuracy, is the right framing for this population.

Most directly relevant of all, Shahi et al. (2024), presented at UIST (the ACM Symposium on User Interface Software and Technology), introduce a vision-based method that lets users design custom gestures with a monocular camera from a single demonstration, using transformers and meta-learning to handle the few-shot problem, and supporting one-handed, two-handed, static, and dynamic gestures. Their user study of twenty custom gestures across twenty-one participants achieved up to ninety-seven percent average recognition accuracy from one demonstration. This paper is the strongest evidence we found that gesture customization from very few examples is both feasible and usable in practice, and it gives us a concrete benchmark to compare our own enrollment accuracy against.

2.6 Stability, Real-Time Behavior, and Evaluation

A recurring theme across this literature is that raw per-frame classification is not enough; a live camera feed produces noisy, flickering predictions that need to be stabilized before they are turned into an action or a spoken word, and this matters even more for an AAC tool, where an accidental trigger produces an unwanted spoken message rather than a harmless media control error.

Molchanov, Yang, Gupta, Kim, Tyree, and Kautz (2016) use a recurrent 3D convolutional network for online detection and classification of dynamic hand gestures, which we reference when discussing our planned extension from static poses to motion-based gestures such as a wave or a circular motion. Tsinganos, Cornelis, Cornelis, Jansen, and Skodras (2022) describe a temporal convolutional network for real-time gesture recognition that assigns a label to a sliding window of predictions only once a threshold is met within that window, otherwise returning a "no gesture" label, a pattern very similar to the majority-vote temporal smoothing already present in our system's design. A related MDPI Bioengineering study on prosthetic hand control also uses a majority-vote window technique to stabilize the prediction stream before it drives a physical action, offering a direct precedent for the majority-vote smoothing block already sketched into our own architecture diagram.

Finally, for evaluating the usability of the enrollment and recognition interface itself, we draw on Nielsen and Molich (1990), whose heuristic evaluation method remains the standard lightweight technique for catching interface problems before a full user study. We plan to run an internal heuristic evaluation pass on our enrollment flow before any hands-on testing with participants.

2.7 Summary of the Gap

Across this literature, we see two threads that rarely meet. One thread (MediaPipe, prototypical networks, few-shot sign language recognition, open set recognition) is strong on the technical mechanisms for learning a new visual category from very few examples. The other thread (AAC foundations, gesture-based assistive interaction) is strong on centering the actual needs of nonverbal and motor-impaired users, but the closest existing systems, particularly Ascari et al.'s work, still rely on retraining a classifier rather than a training-free prototype mechanism, and mainstream few-shot sign language work targets standardized sign vocabularies rather than gestures invented by an individual user. Our project sits at the intersection of these two threads: a training-free, open-set, prototype-based vision pipeline, applied specifically to the problem of letting a nonverbal user define entirely personal gestures without needing an engineer in the loop.

3. Hardware Components Required

Component	Purpose	Notes
Laptop or desktop computer	Runs the recognition pipeline and application	Quad-core CPU minimum, 8 GB RAM minimum (16 GB recommended for training and evaluation), integrated GPU is sufficient given the lightweight embedding network
Webcam	Captures the live hand gesture feed	Built-in laptop webcam or external USB webcam, minimum 720p resolution at 30 fps
Webcam mount or small tripod	Keeps the camera framing stable during 10enrolment and use	Important because inconsistent framing across sessions can hurt recognition consistency
Adjustable desk lighting	Provides consistent, adequate lighting for the vision pipeline	The literature repeatedly notes that vision-based hand tracking is sensitive to lighting conditions, so a simple desk lamp materially improves reliability
Speakers or headphones	Delivers the spoken (text-to-speech) output of a recognized gesture	Any standard laptop speaker or a small external speaker is sufficient
Secondary display or tablet (optional, future work)	Shows a larger, more accessible message board alongside the webcam feed	Not required for the DA1 prototype scope, noted for future accessibility improvements
External microphone (optional)	Not used for gesture recognition, but reserved for a possible future multimodal extension (voice plus gesture disambiguation)	Not required for current scope

4. Software Components Required

Component	Purpose
Python 3.x	Primary development language for the recognition pipeline and application logic
OpenCV	Webcam frame capture, preprocessing, and drawing the live overlay used for enrollment and recognition feedback
MediaPipe (Hands solution)	Real-time extraction of the twenty-one hand landmark keypoints from each frame
PyTorch (or TensorFlow)	Implementation and inference of the embedding network (MLP over landmarks, plus the MobileNetV2-based image stream)
NumPy	Vector math for computing distances between embeddings and averaging examples into prototypes
Text-to-speech engine (e.g., pyttsx3 or a cloud TTS API)	Converts a recognized gesture's associated message into spoken audio output
Simple GUI framework (Tkinter, PyQt, or a lightweight Streamlit interface)	Enrollment interface (guiding the user through providing five to ten examples of a new gesture) and the live recognition/message-board display
Local data storage (JSON, pickle, or SQLite)	Stores each user's profile: enrolled gesture names, prototype vectors, and associated messages
Git and a hosted repository (e.g., GitHub)	Version control and collaboration across the team
Development environment (VS Code or Jupyter Notebook)	Code writing, debugging, and iterative experimentation
LibreOffice or equivalent (for documentation only)	Producing supporting documentation and reports, not part of the runtime system

5. Methodology

5.1 Embedding Backbone (Trained Once, Then Frozen)

To learn a general-purpose space where similar hand shapes land near together and diverse hand shapes land far apart, we first train a tiny embedding network on a wide range of hand poses. This network generates a compact embedding vector from the twenty-one MediaPipe landmark coordinates (plus, if desired, a cropped image of the hand region as a second stream). This network's weights are frozen after it has been trained to a reasonable degree of general separability. The crucial design choice that prevents the catastrophic forgetting issue covered in Section 2.2 is that it is never changed again during regular system operation.

5.2 Enrollment (Few-Shot, Training-Free)

To teach the system a new personal gesture, the user or caregiver selects "enroll a new gesture," names it (and can optionally attach a spoken phrase or on-screen symbol to it), and performs the gesture five to ten times in front of the webcam, ideally from a few slightly different angles. Each demonstration is passed through the frozen embedding network to produce an embedding vector, and these vectors are averaged into a single prototype vector for that gesture. This prototype, along with its associated message, is stored in the user's profile. No gradient descent or retraining happens at this step; it is a single averaging operation, which keeps enrollment fast and keeps previously enrolled gestures completely unaffected.

5.3 Recognition (Live Use)

During live use, each incoming webcam frame is processed as follows: MediaPipe extracts the hand landmarks, the frozen embedding network converts them into an embedding vector, and the system computes the distance from this vector to every stored prototype in the current user's profile. If the closest prototype is within a chosen similarity threshold, that gesture is considered a candidate match; if no prototype is close enough, the frame is labeled as "no gesture," implementing the open-set rejection behavior described in Section 2.2 and preventing an ambiguous hand position from being mistaken for a deliberate gesture.

5.4 Temporal Smoothing

Because a single frame's prediction can flicker due to motion blur, brief occlusion, or a hand passing through an intermediate pose, we apply majority-vote smoothing over a short sliding window of recent frames (for example, the last ten to fifteen frames). A gesture is only confirmed and acted upon (spoken aloud and displayed) once it holds a majority within that window, following the same pattern documented in the temporal stability literature reviewed in Section 2.6.

5.5 Adaptation Loop

If a caregiver or the user notices a gesture being misrecognized or missed, they can flag it, prompting the system to request one additional demonstration and quietly refine only that specific gesture's prototype, again without touching the frozen backbone or any other stored prototype. This turns enrollment from a one-time event into a lightweight, ongoing feedback loop, directly tied to the "puts the user in control" principle from the AAC literature reviewed in Section 2.4.

5.6 Evaluation Plan

We plan to evaluate the system along three lines:

1. **Recognition performance:** per-gesture accuracy, false-trigger rate (how often the system produces a message when none was intended), and the recognition latency from gesture onset to spoken output.
2. **Enrollment usability:** a small internal usability pass using Nielsen's heuristic evaluation method on the enrollment flow, followed by informal testing with a handful of participants (course teammates or volunteers acting as proxy users, given ethical and time constraints on directly testing with the target population at this stage) walking through the process of teaching the system three to five personal gestures.
3. **Stability over time:** confirming that adding new gestures across a session does not degrade recognition of gestures enrolled earlier in that same session, directly testing the claim underlying our choice of a prototype-based rather than fine-tuning-based design.

5.7 Milestones

- DA1 (this submission): topic finalized, literature reviewed, hardware and software requirements identified, methodology defined.
 - DA2 (planned): embedding backbone trained and frozen, enrollment and recognition pipeline implemented end to end.
 - DA3 (planned): temporal smoothing and open-set rejection tuned, text-to-speech and message-board output integrated, internal usability pass completed.
 - Final submission (planned): full system demonstration, evaluation results, and final report.
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References

1. Al Abdullah, B. A., Amoudi, G. A., & Alghamdi, H. S. (2024). Advancements in sign language recognition: A comprehensive review and future prospects. *IEEE Access*, 12, 128871-128895. <https://doi.org/10.1109/ACCESS.2024.3457692>
2. Ascari, R. E. O. S., Pereira, R., & Silva, L. (2019). Personalized interactive gesture recognition assistive technology. In *Proceedings of the 18th Brazilian Symposium on Human Factors in Computing Systems* (pp. 1-12). ACM. <https://doi.org/10.1145/3357155.3358442>
3. Ascari, R. E. O. S., Pereira, R., & Silva, L. (2020). Computer vision-based methodology to improve interaction for people with motor and speech impairment. *ACM Transactions on Accessible Computing*, 13(4), Article 17. <https://doi.org/10.1145/3408300>
4. Bilge, Y. C., Cinbis, R. G., & Ikizler-Cinbis, N. (2023). Towards zero-shot sign language recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(1), 1217-1232. <https://doi.org/10.1109/TPAMI.2022.3143074>

5. Bohacek, M., & Hruz, M. (2023). Learning from what is already out there: Few-shot sign language recognition with online dictionaries. In *2023 IEEE 17th International Conference on Automatic Face and Gesture Recognition (FG)* (pp. 1-6). IEEE. <https://doi.org/10.1109/FG57933.2023.10042544>
6. Colli Alfaro, J. G., & Trejos, A. L. (2022). User-independent hand gesture recognition classification models using sensor fusion. *Sensors*, 22(4), Article 1321. <https://doi.org/10.3390/s22041321>
7. Cui, C., Sunar, M. S., & Su, G. E. (2025). Deep vision-based real-time hand gesture recognition: A review. *PeerJ Computer Science*, 11, e2921. <https://doi.org/10.7717/peerj-cs.2921>
8. Hofman-Zweig, M., & Ben-Assuli, O. (2026). Few-shot sign language recognition using contrastive learning. *Technology and Disability*. <https://doi.org/10.1177/10554181261441139>
9. Kasapbaşı, A., Elbushra, A. E. A., Al-Hardanee, O., & Yilmaz, A. (2022). DeepASLR: A CNN based human computer interface for American Sign Language recognition for hearing-impaired individuals. *Computer Methods and Programs in Biomedicine Update*, 2, 100048. <https://doi.org/10.1016/j.cmpbup.2021.100048>
10. Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D., & Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521-3526. <https://doi.org/10.1073/pnas.1611835114>
11. Koch Fager, S., Fried-Oken, M., Jakobs, T., & Beukelman, D. R. (2019). New and emerging access technologies for adults with complex communication needs and severe motor impairments: State of the science. *Augmentative and Alternative Communication*, 35(1), 13-25. <https://doi.org/10.1080/07434618.2018.1556730>
12. Light, J., & McNaughton, D. (2013). Putting people first: Re-thinking the role of technology in augmentative and alternative communication intervention. *Augmentative and Alternative Communication*, 29(4), 299-309. <https://doi.org/10.3109/07434618.2013.848935>
13. Light, J., & McNaughton, D. (2014). Communicative competence for individuals who require augmentative and alternative communication: A new definition for a new era of communication? *Augmentative and Alternative Communication*, 30(1), 1-18. <https://doi.org/10.3109/07434618.2014.885080>

14. Light, J., McNaughton, D., & Caron, J. (2019). New and emerging AAC technology supports for children with complex communication needs and their communication partners: State of the science and future research directions. *Augmentative and Alternative Communication*, 35(1), 26-41. <https://doi.org/10.1080/07434618.2018.1557251>
15. Light, J., Wilkinson, K. M., Thiessen, A., Beukelman, D. R., & Koch Fager, S. (2019). Designing effective AAC displays for individuals with developmental or acquired disabilities: State of the science and future research directions. *Augmentative and Alternative Communication*, 35(1), 42-55. <https://doi.org/10.1080/07434618.2018.1558283>
16. Mavritsakis, D. (2024). Augmentative and alternative communication in autism spectrum disorder: Transitioning from letter board to iPad, a case study. *Frontiers in Psychiatry*, 15, 1345447. <https://doi.org/10.3389/fpsy.2024.1345447>
17. Molchanov, P., Yang, X., Gupta, S., Kim, K., Tyree, S., & Kautz, J. (2016). Online detection and classification of dynamic hand gestures with recurrent 3D convolutional neural networks. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 4207-4215). IEEE. <https://doi.org/10.1109/CVPR.2016.456>
18. Nielsen, J., & Molich, R. (1990). Heuristic evaluation of user interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 249-256). ACM. <https://doi.org/10.1145/97243.97281>
19. Oudah, M., Al-Naji, A., & Chahl, J. (2020). Hand gesture recognition based on computer vision: A review of techniques. *Journal of Imaging*, 6(8), Article 73. <https://doi.org/10.3390/jimaging6080073>
20. Parisi, G. I., Kemker, R., Part, J. L., Kanan, C., & Wermter, S. (2019). Continual lifelong learning with neural networks: A review. *Neural Networks*, 113, 54-71. <https://doi.org/10.1016/j.neunet.2019.01.012>
21. Rafiq, R. B., Shi, W., & Albert, M. V. (2024). Wearable sensor-based few-shot continual learning on hand gestures for motor-impaired individuals via latent embedding exploitation. *arXiv*. <https://doi.org/10.48550/arXiv.2405.08969>
22. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018). MobileNetV2: Inverted residuals and linear bottlenecks. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 4510-4520). IEEE. <https://doi.org/10.1109/CVPR.2018.00474>
23. Scheirer, W. J., Rocha, A. de R., Sapkota, A., & Boult, T. E. (2013). Toward open set recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(7), 1757-1772. <https://doi.org/10.1109/TPAMI.2012.256>
24. Shahi, S., Mollyn, V., Park, C. T., Kang, R., Liberman, A., Levy, O., Gong, J., Bedri, A., & Laput, G. (2024). Vision-based hand gesture customization from a single

demonstration. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST '24)*. ACM.

<https://doi.org/10.1145/3654777.3676378>

25. Snell, J., Swersky, K., & Zemel, R. (2017). Prototypical networks for few-shot learning. *arXiv*. <https://doi.org/10.48550/arXiv.1703.05175>
26. Tripathi, R., & Verma, B. (2024). Survey on vision-based dynamic hand gesture recognition. *The Visual Computer*. <https://doi.org/10.1007/s00371-023-03160-x>
27. Tsinganos, P., Cornelis, B., Cornelis, J., Jansen, B., & Skodras, A. (2022). Real-time analysis of hand gesture recognition with temporal convolutional networks. *Sensors*, 22(5), Article 1694. <https://doi.org/10.3390/s22051694>
28. Wang, F., Li, C., Zeng, Z., Xu, K., Cheng, S., Liu, Y., & Sun, H. (2021). Cornerstone network with feature extractor: A metric-based few-shot model for Chinese natural sign language. *Applied Intelligence*, 51(10), 7139-7150. <https://doi.org/10.1007/s10489-020-02170-9>
29. Wang, Y., Yao, Q., Kwok, J. T., & Ni, L. M. (2020). Generalizing from a few examples: A survey on few-shot learning. *ACM Computing Surveys*, 53(3), Article 63. <https://doi.org/10.1145/3386252>
30. Zhang, F., Bazarevsky, V., Vakunov, A., Tkachenka, A., Sung, G., Chang, C. L., & Grundmann, M. (2020). MediaPipe hands: On-device real-time hand tracking. *arXiv*. <https://doi.org/10.48550/arXiv.2006.10214>